There are a number of IoT designs that have near identical or identical devices. This means that any security vulnerability found on one device may affect a number of devices that have similar characteristics.

Privacy: The privacy challenge will be formidable in the IoT era if not handled properly. As all devices will be in sync and communicate with each other without human intervention, the risk of privacy leaks will be very high. Devices with voice recognition features can continuously listen to your conversations; and those with vision can watch your activities and then transmit that data to the cloud for processing, which will sometimes involve third parties.

Keeping IoT hardware up-to-date: Regardless of how a company uses the IoT or the cloud, data integrity is a common challenge. At present, the number of devices connected is still not many, but this count will go on increasing exponentially. Then, keeping the entire lot of devices on the same security track, with the same compatibility, will be a huge challenge.

Overcoming connectivity issues: Currently, the IoT uses a centralised, server-client model to provide connectivity to the various servers, workstations and systems. This model is sufficient, as of now, but will it be the same in the future as well? As the IoT becomes more pervasive and billions of devices begin using the network simultaneously, it will just be a matter of time before users start experiencing significant bottlenecks in IoT connectivity, efficiency and overall performance.

IoT application development is currently a task that requires considerable expertise due to its complexity, but it needs to be easier for all developers. There is a need to make hardware and software design simple.

Going forward, the more things get connected to the Internet, the more data there will be created for storage. This large amount of data will need to be cleaned, processed and interpreted.

The desirability of open source in IoT

Here are a few compelling reasons for using open source in IoT:

1. As open source tools are more cost-effective, they are a good choice for companies starting out and facing cost challenges.
2. Open source IoT solutions enable developers to create and change code in their IoT applications to enhance their capabilities and performance.
3. A uniform API is an advantage for developers while using open source frameworks. As the number of sensory devices increases, the code complexity also increases; so the capability to abstract code from one device to another is very important for developers.
4. Open source tools, in comparison to licensed ones, are easier to integrate with other tools.
5. Open source frameworks will improve interconnectivity as they need a central gateway to connect all devices and sensors on the same protocol. For this, software will have to work on low memory and energy as well, because smaller devices will have these kinds of limitations. So apart from specialisation, some features need to be on the same platform and for this interconnectivity, open source frameworks can be very useful.
6. The open source framework approach allows you flexibility. There is the freedom of unlimited customisation and tuning the code to match the specific needs of a product.
7. Open source tools are free from any vendor lock-in as well.
8. More devices connected to the Internet means more data, and open source Big Data tools are very useful for collecting and processing this huge amount of data.

A few open source IoT platforms

1. **Kaa:** This platform is built on a modern cloud-native architecture and has a fully customisable feature set. Complete flexibility, better interconnectivity, data processing customisation and configuration management of device behaviour are some of its key features.
2. **Arduino:** An Arduino development kit offers both software and hardware to developers. Sending messages from one board to another is easy using its cloud based system and its own Arduino programming language. Arduino is well suited for beginners as well, because its software and hardware are designed for those with little or no programming background.
3. **ThingsBoard**: This IoT platform is used for data collection from different sources, and for processing, visualisation and device management. It is rich in functionalities, and has a simple yet powerful API for device connectivity.

The Physical Lab (powered by Google), Eclipse IoT Projects, Open Remote (focused on home automation), Zetta and many other open source IoT platforms are available, catering to diverse user requirements.

The road ahead

As in the case of electricity, automobiles and air planes, which took years from the time of invention to widespread adoption, one of the main challenges to overcome in the IoT world is the notion that everything is fine the way it is. Security and privacy are very critical challenges for IoT. Another challenge is that IoT can be very complex from the technical point of view, and coding and combining all things can be very problematic. This complexity needs a platform that everybody can use, and an open source platform that has the support of the community is the best choice. END

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